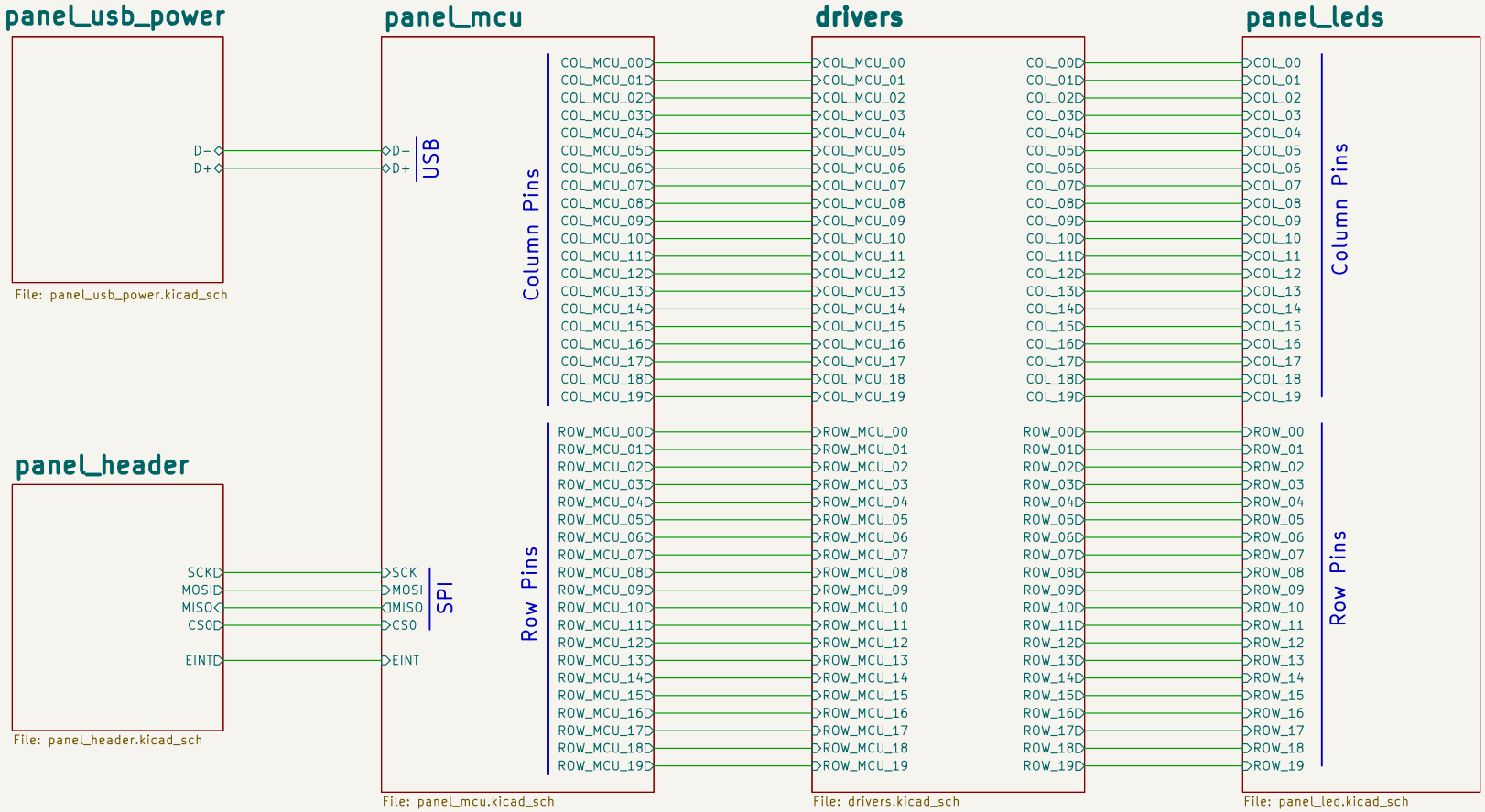


Reiserlab ⊙ Janelia

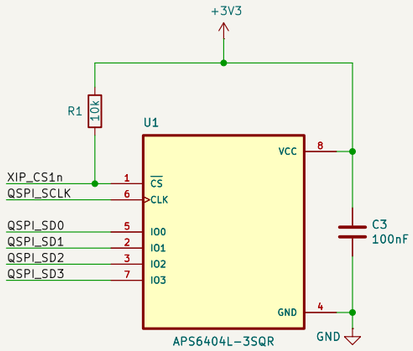
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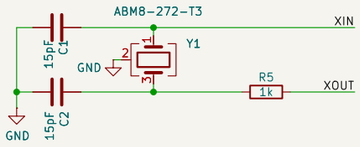
Title: G6 PBYR 45mm RP2354

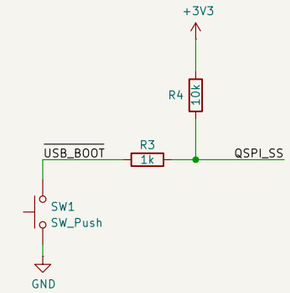
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KiCad E.D.A. 10.0.7		Id: 1/6

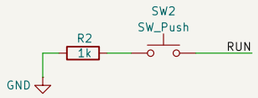
Secondary PSRAM

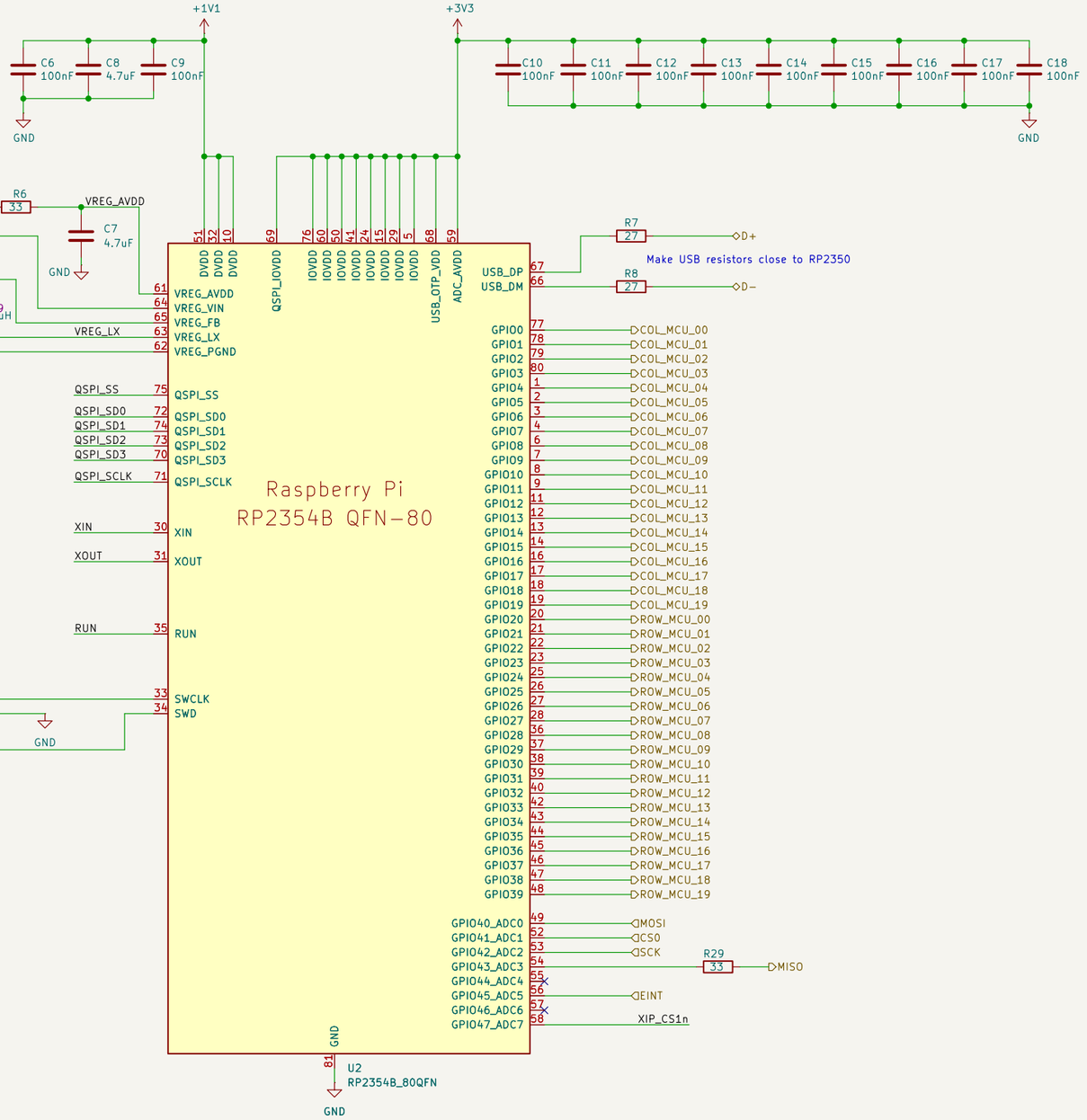


12MHz Crystal









FID1 Fiducial FID2 Fiducial FID3 Fiducial

FID4 Fiducial FID5 Fiducial FID6 Fiducial

PWR_FLAG +1V1 PWR_FLAG VREG_AVDD

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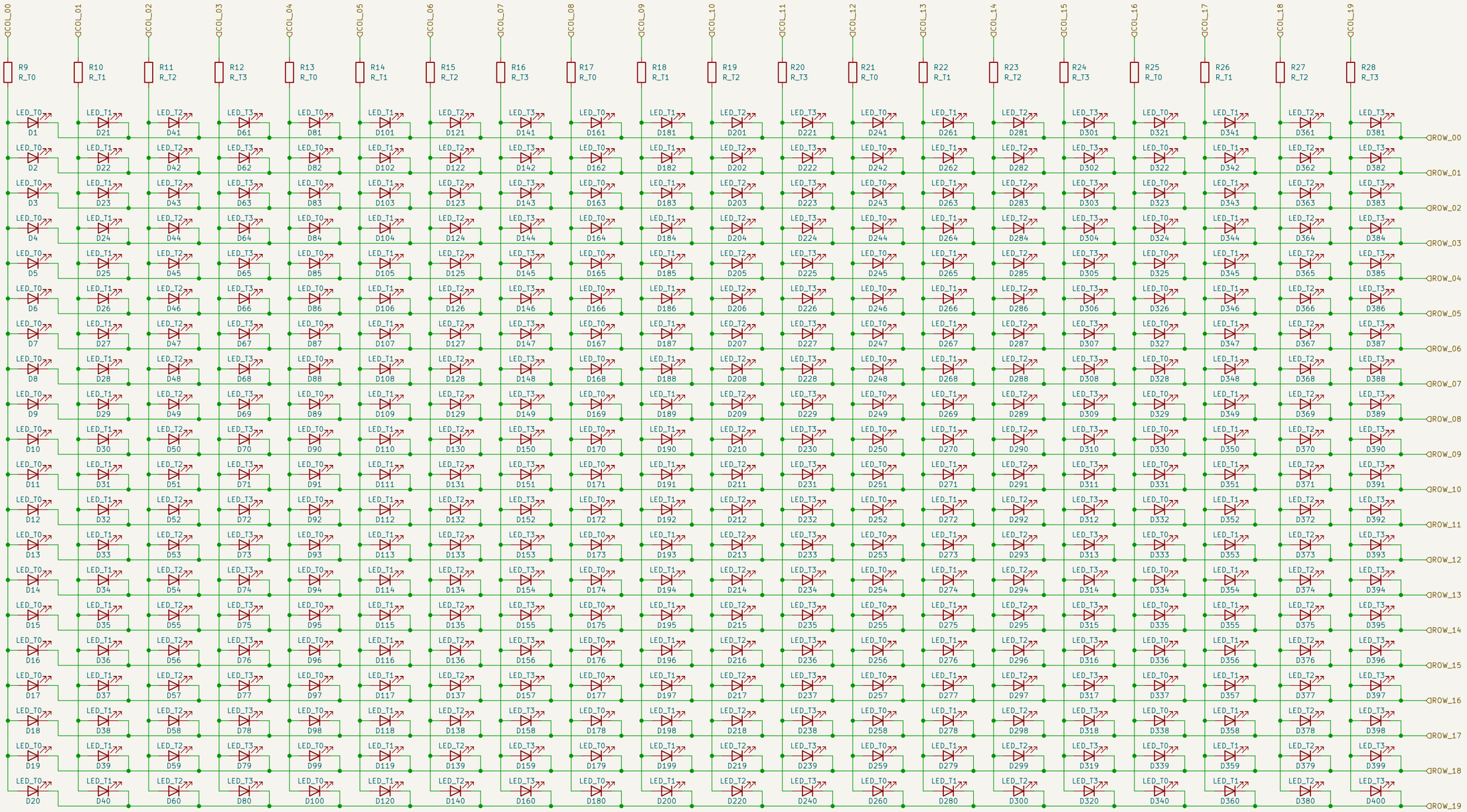
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Title: G6 PBYR 45mm RP2354

Size: User | Date: 2026-10-26 | **Rev: v0.5.2**

KiCad E.D.A. 10.0.7 | Id: 2/6



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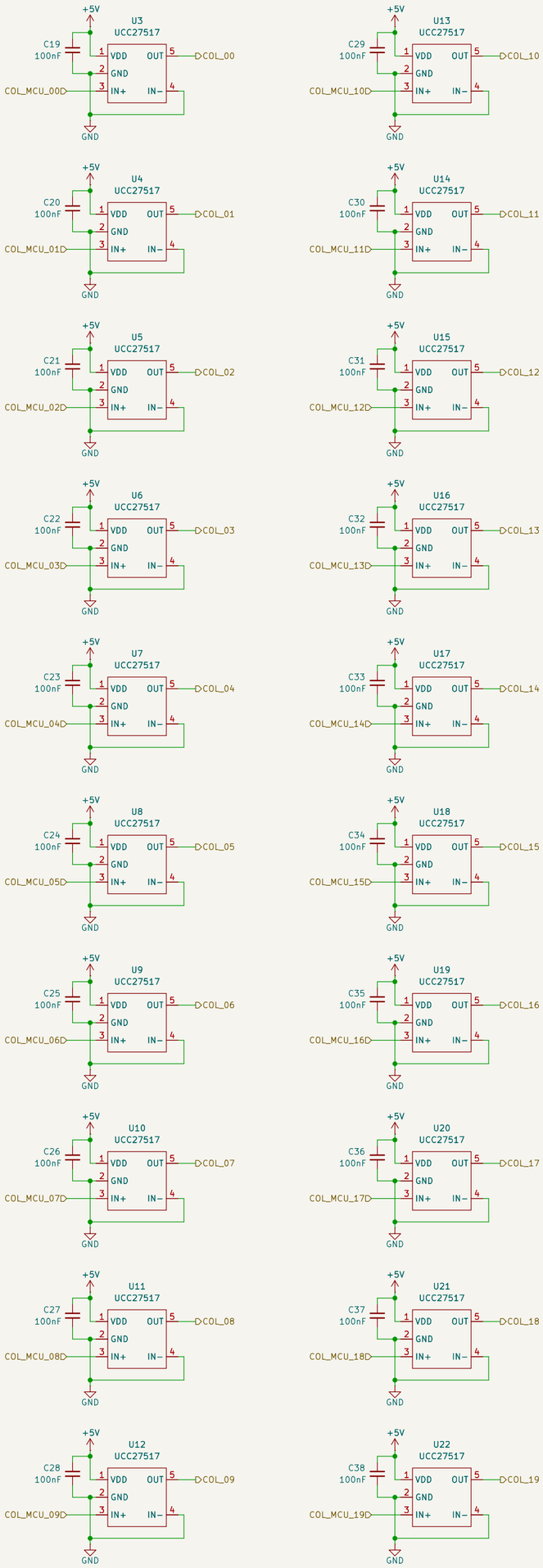
Sheet: /panel_leds/

File: panel_led.kicad_sch

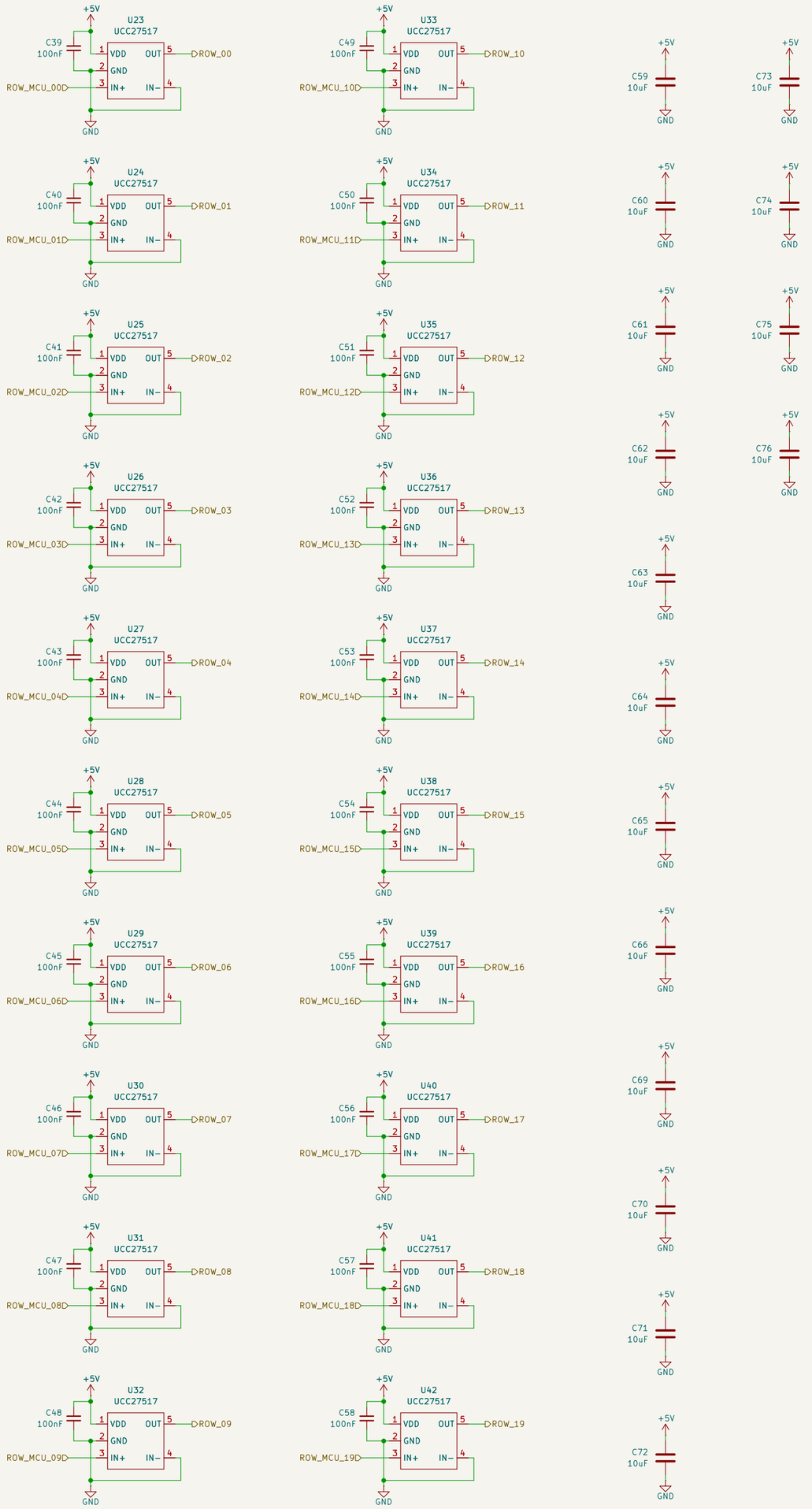
Title: G6 PBYR 45mm RP2354

Size: User	Date: 2026-10-26	**Rev: v0.5.2**
KiCad E.D.A. 10.0.7		Id: 4/6

Column Drivers



Row Drivers

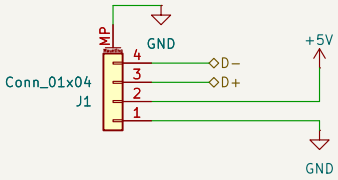


Reiserlab @ Janelia
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File: drivers.kicad_sch

Title: G6 PBYR 45mm RP2354

Size: User	Date: 2026-10-26	**Rev: v0.5.2**
KiCad E.D.A. 10.0.7		Id: 6/6

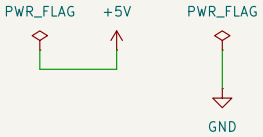
USB Connection
JST-SH QWIIC/STEMMA-QT style connector



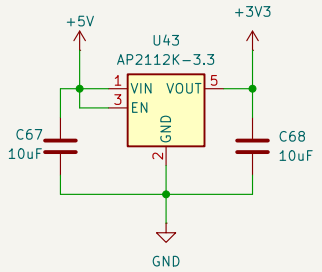
For connecting cable

USB to Grove
https://shop.m5stack.com/products/connector-grove-to-usb-c

Grove to JST-SH
https://www.sparkfun.com/products/15109



3.3V Regulator



3.3V Regulator
Manufacturer PN: AP2112K-3.3TRG1
LCSC PN: C51118
Digikey PN: AP2112K-3.3TRG1DICT-ND

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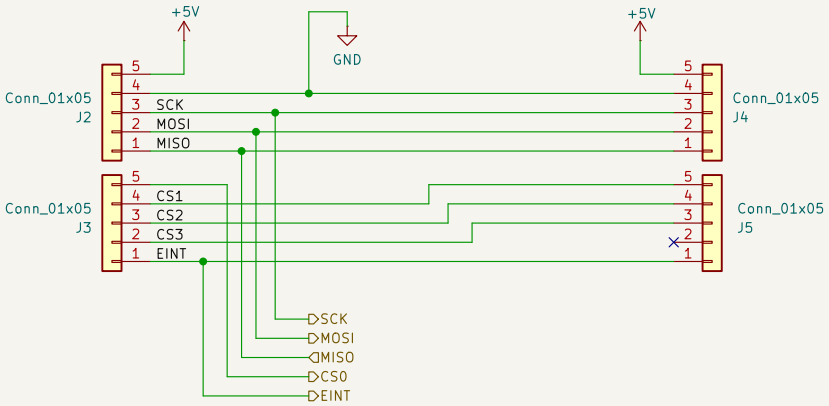
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File: panel_usb_power.kicad_sch

Title: G6 PBYR 45mm RP2354

Size: A4	Date: 2026-10-26	**Rev: v0.5.2**
KiCad E.D.A. 10.0.7		Id: 7/6

Bottom Headers

Top Headers



1x5 Pin Header

JLCPCB/LCSC
PN C7296801

Digikey PN: 952-M20-8890545RCT-ND
Manufacturer PN: M20-8890545R

Digikey PN: 952-3266-ND
Manufacturer PN: M20-8900505

LCSC for v0.1.3: C46061678
Manufacturer PN: HX PZ2.54-1x5P WT

1x5 Female Header
Digikey PN: SMH-105-02-T-S
Manufacturer PN: SMH-105-02-T-S

LCSC for v0.1.3: C5142238
Manufacturer PN: X6511FRS-05-C85D30

Reiserlab @ Janelia

Sheet: /panel_header/
File: panel_header.kicad_sch

Title: G6 PBYR 45mm RP2354

Size: A4	Date: 2026-10-26	**Rev: v0.5.2**
KiCad E.D.A. 10.0.7		Id: 8/6